

Dinusha Ekanayake

Intern AI/ML Engineer

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Moratuwa, Sri Lanka | LinkedIn Profile

GitHub Profile | Portfolio Website

Professional Summary

Motivated third year B.Sc. (Hons.) Artificial Intelligence undergraduate at the University of Moratuwa with a strong interest in machine learning, deep learning, and applied AI. Skilled in building and fine tuning machine learning models, working with large language models and AI agents, and developing full stack AI applications. Passionate about turning data into practical solutions, learning new tools quickly, and building AI products that solve real problems. A dedicated and fast learner with good teamwork, communication, analytical, and problem solving skills.

Education

B.Sc. (Hons.) in Artificial Intelligence **2024 – Present**

University of Moratuwa *Moratuwa, Sri Lanka*

- CGPA: 3.58 / 4.00

G.C.E. Advanced Level Examination **2020 – 2023**

S. Thomas' College Bandarawela *Bandarawela, Sri Lanka*

- Common Stream Badulla District 2nd | Z-Score: 1.9214 (Combined Mathematics: A, Physics: B, ICT: A)

Skills

- **AI, Machine Learning & LLM Engineering:** PyTorch, TensorFlow, Scikit-learn, Transformers, Hugging Face, Kaggle, LangChain, LangGraph, RAG, SHAP, XGBoost, LightGBM, CatBoost, DistilBERT, DistilBART
- **Data Science & Visualization:** NumPy, Pandas, Matplotlib, Seaborn, Google Colab, Jupyter Notebook
- **Programming Languages:** Python, Java, JavaScript, TypeScript, C, C++, PHP, and Prolog
- **Frameworks & APIs:** FastAPI, Node.js, Express.js, React.js, Next.js, Tailwind CSS
- **Databases & Cloud:** PostgreSQL, MySQL, Supabase, Firebase, AWS EC2, Vercel, Render
- **Developer Tools:** Git, Postman, Thunder Client, Cisco Packet Tracer, Jira, Figma
- **Design & Soft Skills:** Photoshop, Lightroom, Canva, Leadership, Project Management, Communication, Teamwork, Analytical Thinking, Problem Solving
- **Languages:** English (Fluent), Sinhala (Native)

Key Projects

PredictiX | GitHub | Demo **Second Year Industry Based AI Software Project**

AI Predictive Maintenance Platform ***(In collaboration with Zone24x7)***

Tech Stack: Python, FastAPI, Next.js, TypeScript, PostgreSQL, Supabase, AWS EC2, Vercel, XGBoost, LightGBM, CatBoost, DistilBERT, DistilBART, LangChain, Groq and SHAP

- Engineered an end to end AI powered predictive maintenance platform for warehouse vehicle fleets, integrating predictive analytics, intelligent ticket management, asset intelligence and AI assisted reporting.
- Built machine learning models using LightGBM, XGBoost and CatBoost for failure prediction, maintenance forecasting and maintenance cost estimation, with SHAP explanations for model transparency.
- Developed a Weibull survival analysis model using Lifelines to estimate component survival probabilities and expected remaining lifetime.
- Developed ticket categorization and priority prediction models by fine tuning DistilBERT and used DistilBART for ticket and asset summarization.
- Built RAG pipelines using LangChain and Groq Llama 3.3 for AI generated warehouse and asset reports, and developed a role based agentic chatbot with intelligent routing and database interaction.
- Developed and deployed the system using a Next.js and TypeScript frontend on Vercel, a FastAPI backend on AWS EC2 and a PostgreSQL database hosted on Supabase.

Multi Agent Research Assistant | GitHub **Individual Project**

AI Powered Research, Fact Checking and Report Generation System

Tech Stack: Python, LangChain, OpenAI, Pydantic, Streamlit and DOCX

- Built a multi agent research assistant in which three LLM agents research a topic, fact check claims and generate a structured report.
- Used LangChain and Pydantic schemas to create reliable, structured handoffs between the agents.
- Added web search as a callable tool and implemented one click DOCX report generation through a Streamlit interface.

CV Mate | GitHub | Demo **Individual Project**

AI Resume Builder

Tech Stack: React, Vite, Firebase, Gemini API, Electron, PDF, DOCX and JSON

- Built a browser based CV builder with live preview, ATS scoring and keyword gap detection.
- Integrated Gemini API for profile summary generation, bullet point rewriting and job description matching.
- Added Firebase cloud synchronization and export support for PDF, DOCX and JSON formats.
- Developed an Electron desktop version of the application.

ReClaim | GitHub **Individual Project** | Ongoing

Smart Lost and Found Platform

Tech Stack: Next.js, Node.js, Express.js, PostgreSQL, Prisma, OpenAI Embeddings, Socket.io, Cloudinary and JWT

- Developed a smart lost and found platform that matches lost and found reports using category, keywords, GPS, date and semantic similarity.
- Integrated OpenAI embeddings to improve item matching beyond conventional keyword based methods.
- Implemented JWT authentication, real time chat using Socket.io, Cloudinary image uploads and an administrative dashboard.
- Deployed the frontend on Vercel and the backend services on Render.

SmartFlow | GitHub **First Year Hardware Project**

Smart People Tracking Fan

Tech Stack: ESP32, ESPHome, HLK-LD2450 mmWave Radar, DHT22, Servo Motor, AC Dimmer and OLED

- Built an ESP32 based smart fan that detects people using an mmWave radar sensor and rotates a servo to follow their position.
- Enabled tracking of up to three targets using their X and Y coordinates.
- Read room temperature and humidity using a DHT22 sensor and adjusted fan speed through an AC dimmer.
- Added physical button controls, an OLED display and remote Wi-Fi monitoring using ESPHome.

Achievements

- **Finalist - DataXplore 2.0** | Data Science Competition organized by the University of Sri Jayewardenepura
- **Best G.C.E. Advanced Level Results, Common Stream** | S. Thomas' College Bandarawela
- **Best Pillar Member of the Term 2024/25, Photography Pillar** | MoraSpirit, University of Moratuwa

Courses & Certifications

- Machine Learning Specialization | DeepLearning.AI
- Introduction to Cybersecurity | Cisco
- Python Programming and Frontend Web Development | University of Moratuwa
- Intermediate Machine Learning and Time Series | Kaggle
- Python and Data Science | University of Colombo

Leadership and Activities

Head of Photography Pillar **2024 – Present**

MoraSpirit, University of Moratuwa

Web Developer **2025 – Present**

IoT and Embedded Systems Research Labs, Faculty of Information Technology, University of Moratuwa

Assistant Head of Photography and Videography Pillar **2024 – Present**

FIT Moments, Faculty of Information Technology, University of Moratuwa

References

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